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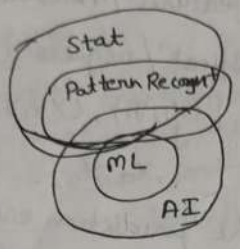
		Data Science & Machine Learning Revision		

* "Learning is any process by which a system improves its performance from experience." (data)

- Alexander Simon
(Nobel Prize in Economics & Turing Award Winner)

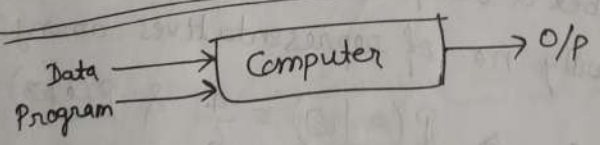
*



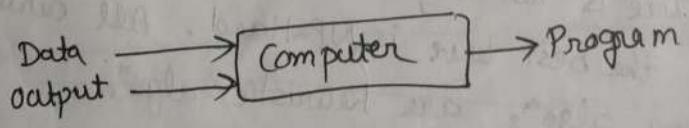
* ML is a field of Computer science that uses statistical techniques to give computer systems the ability to "learn" (i.e. progressively improve performance on a specific task) with data, w/o being explicitly programmed.

* The name "ML" was coined by Arthur Samuel.

* Traditional Programming



Machine Learning



* ML → Supervised Learning (Task Driven) < Classificat, Regression
→ Unsupervised " (Data Driven) < Clustering, Dimensionality Red.
→ Reinforcement " (Learn from Mistakes)

* Data records / examples / instances / cases / samples / observations
attributes / variables / features / measurements / dimensions

Goal To learn a classificat^n / regression model "f"
from the data $\{(x_1, y_1), (x_2, y_2), \ldots\}$
$\downarrow$
$(x_{11}, x_{12}, x_{13}, \ldots)$

by minimizing the prediction error / Loss funct^n
on training set.

& Apply "f" to a never seen example x & O/P
the predicted value $y = f(x)$.

* Classifiers ⇒ KNN, SVM, Decision Tree, Neural N/w, Naive Bayes, Bayesian N/w, Logistic Regression, Random Forest, Deep Learning N/w etc

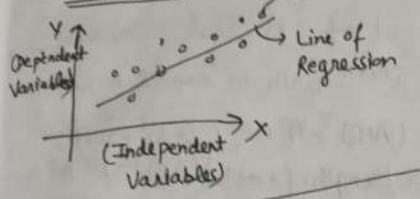
* "No Free Lunch Theorem in statistics" - David Wolpert.

* KNN = It label a test point x to that class that has a majority no. of representatives among the neighbors of x.

$\Rightarrow P(\circ \mid \boxtimes) = \frac{4}{6}$ & $P(\bullet \mid \boxtimes) = \frac{2}{6}$

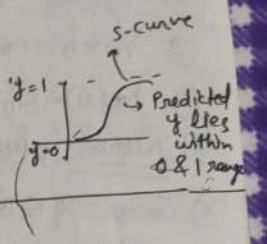
* Decision Tree is not unique. We want smaller & accurate tree. Finding the best tree is NP-Hard. All current tree building algo^{n}s are heuristic algo^{n}s.

(*) Regression



Types of Regression

Linear Regⁿ,
Polynomial "
SVM "
Decision Tree "
Random Forest "
Ridge "
Lasso "
Logistic "



(*) Reinforcement Learning

Training info: Evaluations. (rewards/penalty)

I/p → [Reinforcement Learning] → o/p = actions

(*) Deep Learning is the learning of representatⁿs of in data. If you provide the system tons of infom, it begins to understand it & respond in useful ways.

(*) ML: I/p → Feature Extractⁿ → Classificatⁿ → o/p.

DL: I/p → [Feature Extractⁿ & Classificatⁿ] → o/p.

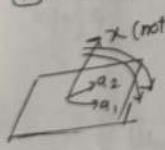
(*) 2 DL Inference Models
- → Discriminative Models → P(y|x) → Regressions/SVMs.
- → Generative Models. → Estimate P(x|y) & then find P(y|x)
  - GAN, Naive Bayes

(*) [Training Data] —Training→ [Generative Model] —sampling→ [Generated Sample]

Random Noise

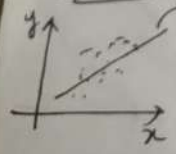
Matrix Algebra & Calculus.

- Vector in $\mathbb{R}^n$ is an ordered set of 'n' real numbers.
- $m \times n$ matrix is an object in $\mathbb{R}^{m \times n}$ with m rows & n columns.
- $tr(A) = A_{11} + A_{22} + \dots$; $(A^T)^T = A$; $(A+B)^T = A^T + B^T$; $(rA)^T = rA^T$; $(AB)^T = B^T A^T$
- Norms quantify "size" of a vector / length of a vector
- Given $x \in \mathbb{R}^n \Rightarrow \|cx\| = |c|\,\|x\|$ & $\|x\| = 0 \Leftrightarrow x = 0$ & $\|x+y\| \le \|x\| + \|y\|$
- Euclidean L_2-norm $\Rightarrow \|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots x_n^2}$
  L_1-norm (Manhattan dist) $\Rightarrow \|x\|_1 = |x_1| + |x_2| + \dots + |x_n|$
  L_∞-norm (Uniform Norm) $\Rightarrow \|x\|_\infty = \max_i |x_i| \Rightarrow \|(1, -1, 5)\|_\infty = 5$
- Norms can also be defined for matrices.
  ex. Frobenius Norm:- $\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}^2} = \sqrt{tr(A^T A)}$
- $\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cos\theta$.
- $x^T y = \sum_{i=1}^{n} x_i y_i$
- $rank(xy^T) = 1$ (vector outer prod)
- If $u \cdot v = 0$; $\|u\|_2 \,!= 0$; $\|v\|_2 \,!= 0 \Rightarrow u$ & v are orthogonal
  If $u \cdot v = 0$; $\|u\|_2 = 1$; $\|v\|_2 = 1 \Rightarrow u$ & v are orthonormal
- When we do $y = Ax \Rightarrow$ matrix A boils down a vector x to its column space



$\Rightarrow$ When we do Ax, then matrix A rotates x & comes it into plane.
$\Rightarrow$ Certain vectors are there for which matrix can't rotate. They can only stretch those vectors. $\Rightarrow$ Eigen Vectors
$\Rightarrow$ Eigen Vectors are rotational invariant. $Ax_e = y = \lambda x_e$.

Regression Probm



We need m & c to find this best fit line. $y = mx + c$.
Suppose there are 3 points on this line, then.

$$\left.\begin{aligned} mx_1 + c &= y_1 \\ mx_2 + c &= y_2 \\ mx_3 + c &= y_3 \end{aligned}\right\} \text{Overdetermined system.} \Rightarrow \underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} m \\ c \end{bmatrix}}_{z} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{b}$$

$\rightarrow Az = b$

Since, system is not fully solvable so, b is not in Col space of A

But we can find relaxed/approximate version of the probm. Make Projectn of $\vec{b}$ in colm space of A say $\hat{b}$ & so, $Az^* = \hat{b}$ & $e = b - \hat{b}$ & $A^T e = \vec{0} \Rightarrow A^T(b - \hat{b}) = 0 \Rightarrow A^T b = A^T \hat{b}$
$\Rightarrow A^T b = A^T A z^* \Rightarrow \boxed{z^* = (A^T A)^{-1} A^T b}$ & Projectn Matrix $Pb = \hat{b} = Az^* = A(A^T A)^{-1} A^T b$
$\Rightarrow \boxed{P = A(A^T A)^{-1} A^T} \Rightarrow \boxed{P^2 = P}$

- OLS $\Rightarrow$
- Positive
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  Gradient $\Rightarrow \nabla f$
- Level Curves $\Rightarrow$
  Ex. x_1^2
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$= x A^T$ ∴
$= B^T A^T$

$\|x\| \le \|x\| + \|y\|$

$-1, 5 \|_\infty = 5$

vector outer product

$y^T) = 1$

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$A^T b$

- OLS $\Rightarrow \|b - Az\|_2^2 = \sum (y_i - \hat{y}_i)^2$
- Positive Semidefinite $A \in \mathbb{R}^{m \times m}$ iff $x^T A x \geq 0 \quad \forall x \in \mathbb{R}^m$
  & if $x^T A x > 0$, Then A = +ve definite
- Orthogonal $U \in \mathbb{R}^{m \times m}$ iff $U^T U = U U^T = I$
  Each row/column of an orthogonal matrix is orthonormal ~~matrix~~ vector
- If A is +ve definite then all eigen values = +ve.
- Subspace $V \in R^n$ satisfies $0 \in V$ & if $x, y \in V$ & $c \in \mathbb{R}$ then $c(x+y) \in V$.
- vectors $x_1, x_2, \ldots, x_m$ span V if $V = \{ \sum \alpha_i x_i \mid \alpha \in \mathbb{R}^m \}$.
- 4 Fundamental subspaces for matrix M. } Null(M) ⊥ row(M).
  Col(M), Row space, Null, (M), left-Null. } $\underbrace{\dim(\text{Null}(M))}_{\text{nullity}} + \underbrace{\dim(\text{row}(M))}_{\text{rank}} = n$
- rank(M) = dimensionality of row/col spaces.
- If $\underbrace{\text{rank}(M) = k}_{M_k^{m \times n}}$ then $m\{\boxed{M}_{n}\} = m\{\boxed{M}_{k}\}\{\underbrace{\quad}_{n}\}k$ } Rank Factorizat (used in Recommen-der systems)
- $M^{-1} M = M M^{-1} = I$; $(M^{-1})^{-1} = M$; $(M^T)^{-1} = (M^{-1})^T$; $(MA)^{-1} = A^{-1} M^{-1}$
- $Mx = y \Rightarrow$ sol^n exists if $y \in$ Col(M) ; Possibly ∞ no. of sol^{n}s
- The term eigenvalue is from German word Eigenwert (Proper value)
- Eigen Decomposition $\Rightarrow A = V \Sigma V^{-1}$ (V = Matrix from Eigenvectors of A)
  ↳ Exist if eigenvectors are linearly independent
- If M = Real Symmetric then Q is orthogonal & $A = Q \Sigma Q^T$ $(Q^{-1} = Q^T)$
  ↳ $QQ^T = I \Rightarrow Q^{-1} = Q^T$
- If $A_k^{m \times n}$ is not square matrix then it is SVD $\Rightarrow A = U \Sigma V^T$

- Matrix Calculus

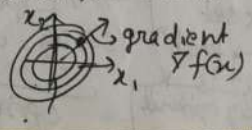
$f: \mathbb{R}^n \to R \Rightarrow$ Ex. $f(x) = x_1^2 + x_2^2$

Gradient $\Rightarrow \nabla f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \end{bmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \end{pmatrix}$

Level Curves ⇒ Curves for which f(x) = Constant

Ex. $x_1^2 + x_2^2 = k$

↳ Contour maps = Const. Funct^n curve.



→ Gradient is always normal to the level curve & it points in the direct^n in which funct^n is increasing very fast

→ $\nabla(x^T a) = a$ → $\nabla(x^T A x) = Ax + A^T x$

→ 1st order optimality cond^n $f'(x^*) = 0$

→ In multidimensional system, gradient should by vanished $\nabla f(x^*) = 0$
(x^* = critical point = local min/max/saddle point)

→ $\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$ = Hessian Matrix
(2nd order optimality)

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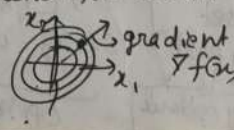

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(*) Optimization (Largely unsolved)

→ It is all about minimizat^n / maximizat^n.

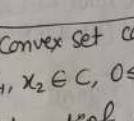
→ Most real world prob^{m}s are non-convex in nature

→ When there are multiple local minima & local maxima then these funct^{n}s are called multimodal funct^n.

→ Only one minima ⇒ Such funct^{n}s are called Convex funct^n

→ If $H \succ 0 \Rightarrow x^*$ is minimum & if $H \prec 0 \Rightarrow x^*$ is maximum.

→ Convex Funct^n ⇒ $x_3 = \theta x_1 + (1-\theta)x_2$; $0 \le \theta \le 1$.



⇒ Def^n of Convex Fun ⇒ $\theta f(x_1) + (1-\theta) f(x_2) \ge f(\theta x_1 + (1-\theta)x_2)$ $\left(\begin{array}{l}\forall \theta \in [0,1] \\ \forall x_1 x_2 \in \text{dom}\end{array}\right)$

⇒ In DL, most objective fun^s are non-convex

(*) Convex set contains line segment b/w any 2 points in the set.

$x_1, x_2 \in C,\ 0 \le \theta \le 1 \Rightarrow \theta x_1 + (1-\theta)x_2 \in C$.

→ Intersect^n of convex sets is convex. f = convex ⇒ $-f$ concave & vice-versa

→ e^x, $-\log x$, $x \log x$ are convex.

→ Steepest Descent Method ⇒ $x^{k+1} = x^k \pm \alpha^k \nabla f(x^k)$ [k = iterations]

↳ Gradient is the rate of change of funct^n at a point. Using gradient as the search direct^n helps reduce no. of iterations needed

↳ For convergence → $|f(x^{k+1}) - f(x^k)| < \varepsilon_1$ (tolerance)

↳ Initialize $x^0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ → find $\nabla f(x^0)$ → find $x^1 = \underbrace{x^0 \pm \alpha^0 \nabla f(x^0)}_{\text{func}^n \text{ of } \alpha^0}$

↓ find $f(x^1)$ → $\frac{d f(x^1)}{d\alpha^0} = 0$ & get α^0 & get x^1 & repeat

→ A Quadratic funct^n = $x^T A x$

→ In Newton's method for optimizat^n, we use, $x^{k+1} = x^k - \underbrace{[\nabla^2 f(x^k)]^{-1}}_{\alpha^k = \text{Hessian}} \nabla f(x^k)$

↳ It reduces no. of iterat^{n}s but increases computat^n

(*) Prob^m Min $f(x)$ subject to $h(x) = c$, $g(x) \ge b$ & $x \in S$

if $f(x)$ is a convex funct^n & S = convex set

Then it is called Convex Programming prob^m (locally optimal sol^n is globally optimal)

$\nabla f = \lambda \nabla g$

$\Rightarrow \nabla [f(x,y) - \lambda g(x,y)] = 0$

$\Rightarrow \nabla L(x, y, \lambda) = 0$

So, $\frac{\partial L}{\partial x} = 0$, $\frac{\partial L}{\partial y} = 0$, $\frac{\partial L}{\partial \lambda} = 0$

Optimality cond^n (KKT)

(*) SVM

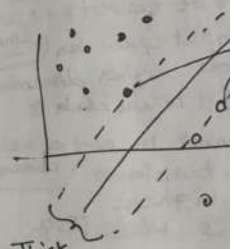
→ They are also

→ Invented by

→ They are separate

→ Kernel func

→ Here prob



Thick margin (For Thin margin, Error will be high)

↳ So maximizi

↳ So, Prob^m m

By using Lagran

→ Primal SVM Pr

→ SVM follows s

strong duality (u

holds for convex o

→ Hard Margin S

Soft "

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funct^n.
ex funct^n
is maximum.

$+(1-\theta)x_2)$
$+\ \theta \in [0,1]$
$x_1, x_2 \in$ dom f
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& vice-versa

[K = iterations

Using
ns needed

$f(x^0)$
$\bar{x}^0$
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1 &
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0

KT)

⊛ SVM

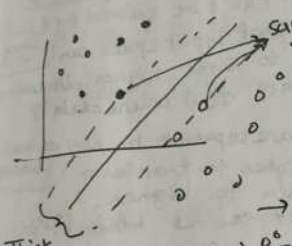
→ They are also called strong learners.

→ Invented by V. Vapnik in Russia in 1970s.

→ They are linear classifiers & finds a hyperplane to separate 2 classes of data +ve & −ve.

→ Kernel funct^n's are used for non-linear separation.

→ Here prob^m is convex optimizat^n & guaranteed to be solved

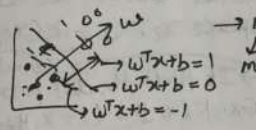


→ Define margin of a linear classifier as the width that the boundary could be increased by before hitting a datapoint.

→ Max^m Margin classifier is with max^m margin

→ Support vectors are those data points that the margin pushes up against.

→ We should have to maximize the margin.



→ $m = \frac{2}{\|w\|_2}$ ($\because \left|\frac{b-1-b+1}{\|w\|_2}\right| = \frac{2}{\|w\|_2}$)

↳ margin

(For Thin margin, Error chance will be high)

↳ So maximizing $\frac{2}{\|w\|_2}$ is same as minimizing $\underbrace{\frac{1}{2}\|w\|_2^2}_{\text{Convex funct}^n}$

For class ① $w^Tx_i+b \geq 1$ & $y_i = 1$

" " ② $w^Tx+b \leq -1$ & $y_i = -1$

↳ So, Prob^m. $\underset{w,b}{\text{minimize}}\ \frac{1}{2}\|w\|^2 = \frac{1}{2}w^Tw$ such that $y_i(w^Tx_i+b) \geq 1$

By using Lagrangian multiplier: $L(w,b,\lambda) = \frac{1}{2}w^Tw - \sum_{i=1}^{n}\lambda_i(y_i(w^Tx_i+b)-1)$

→ Primal SVM Prob^m. $\underset{w,b}{\min}\ \underset{\lambda}{\max}\ L(w,b,\lambda)$ $\quad x_1, x_2 \to w^Tx \Rightarrow [w_1, w_2]\begin{bmatrix}x_1\\x_2\end{bmatrix}$

→ SVM follows strong Duality relat^n (Primal, Dual) → $\lambda = \alpha$ (Lagrangian multiplier) $\neq 0$ for support vectors

strong duality (weak duality $d < b$) holds for convex optimizat^n prob^m with constraints.

⊛ Wolfe dual Prob^m

$\underset{\alpha}{\text{maximize}} \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y_i y_j x_i x_j$

depends only on α (Lagr. multiplier.

such that $\alpha_i \geq 0$ & $\sum_{i=1}^{m}\alpha_i y_i = 0$

→ Hard Margin SVM = Linearly Separable data

Soft " " = Non-" " "

$w = \sum_i \alpha_i y_i x_i$ & $b = y_i - w_i x_i$

So, we get the hyperplane

⇒ Non-linearly separable Prob^m

↳ We allow "error" ξ_i in classificat^n based on o/p of discriminant fun. $w^T x + b$.

↳ approximates the no. of misclassified points

↳ New objective fun. $\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i$ (C = tradeoff hyperparameter b/w error & margin

large C = high penalty for errors

s.t. $y_i(w \cdot x_i + b) \geq 1 - \xi_i$ & $\xi_i \geq 0$.

↳ slack variable. Regularizat^n P.ramete

Wolf-dual Prob^m

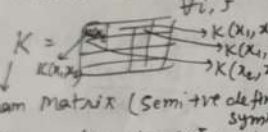
$\max W(\alpha) = \sum_i \alpha_i - \frac{1}{2}\sum_i \sum_j \alpha_i \alpha_j y_i y_j \, x_i \cdot x_j$

s.t. $C \geq \alpha_i \geq 0$

$\sum \alpha_i y_i = 0$

⇒ Every semi-positive definite symmetric funct^n is a kernel (Mercer's Theorem)

↳ $K(x_i, x_j) = K(x_j, x_i)$ $\forall i, j$

$K =$ 

Gram Matrix (Semi +ve definite symmetric)

(*) Polynomial kernel $= (x^T y + 1)^d$ (d = degree)

(*) RBF kernel $= \exp(-\|x - y\|^2 / 2\sigma^2)$

(*) $K(x, y) = \tanh(k x^T y + \theta)$. (~ = sigmoid fn)

(*) Non-Linear SVMs (Kernel Machines)

↳ Ex. XOR prob^m can't be solved by LSVM (Linear SVM)

↳ Idea: The original input space can be mapped to some higher dimensional feature space $\phi(x)$ where data is separable.

↳ N data points are separable in N-1 or more dimens.

↳ Comput^n is higher to transform low dimension to higher.

↳ Kernel tricks solves this issue.

↳ Replace $x_i \cdot x_j = \underbrace{K(x_i, x_j)}_{\text{Kernel funct}^n} = \phi(x_i) \cdot \phi(x_j)$

→ It finds the dot product in higher dimension

Ex. $K(x, y) = (1 + x_1 y_1 + x_2 y_2)^2$ (we don't need ϕ)

(*) Multiclass classificat^n → One-against-all

→ " " one $(K(K-1)/2)$ classifier (K = class.

Use Majority Voting in classificat^n

(*) KNN does not build model from training data.

↳ It is parametric & weak learner

↳ SVM is strong learner bcoz one decision boundary is fixed, we are done

↳ decision tree is also weak learner

(*) Decision Tree Classifiers.

↳ A greedy divide & Conquer algo^m. Tree is constructed in a Top down recursive manner.

↳ Attributes are selected on the basis of an impurity funct^n (e.g. information gain).

↳ The objective is to reduce impurity (mixing of class labels) or uncertainty in data as much as possible

↳ A subset of data is pure if all instances belong to the same class.

↳ Attributes (based

↳ Entropy

↳ Entropy (

↓

Impurity Measure.

↳ 100 % +ve

↳ As data

↳ gain (D

↳ Handle (can be we can u

↳ Overfitti

↳ No classifier

↳ Complex Mo Simple

(*) Variance: of model or a value

⇒ Error = (Test)

↳ B

⇒ Ensemble

↳ The ensem classifi